

# EXPERIMENT 19

## TRIGGER

### Aim:

To implement a trigger in MySQL.

### Description:

A trigger is a stored program that executes automatically in response to events such as INSERT, UPDATE, or DELETE.

### Syntax-Create trigger

```
CREATE TRIGGER trigger_name trigger_time trigger_event
```

```
ON table_name
```

```
FOR EACH ROW
```

```
BEGIN
```

```
...
```

```
END;
```

### Question 1

The manual asks to create a **BEFORE UPDATE trigger** that logs changes into an audit table.

## OUTPUT:

1)

```
CREATE TABLE Patient_Audit (  
    Audit_ID INT AUTO_INCREMENT PRIMARY KEY,  
    Patient_ID VARCHAR(10),  
    Old_Patient_Name VARCHAR(100),  
    New_Patient_Name VARCHAR(100),  
    Updated_At TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
DELIMITER //  
  
CREATE TRIGGER Patient_Update_Audit  
BEFORE UPDATE ON Patient  
FOR EACH ROW  
BEGIN  
    INSERT INTO Patient_Audit  
    (  
        Patient_ID,  
        Old_Patient_Name,  
        New_Patient_Name  
    )  
    VALUES  
    (  
        OLD.Patient_ID,  
        OLD.Patient_Name,  
        NEW.Patient_Name  
    );  
END  
  
35 DELIMITER ;  
36  
37 • UPDATE Patient  
38 SET Patient_Name = 'Arun Kumar'  
39 WHERE Patient_ID = 'P001';  
40  
41 • SELECT * FROM Patient_Audit;
```

| Result Grid     Filter Rows: <input type="text"/>   Edit:      Export/Import:     Wrap Cells |            |                  |                  |                     |
|----------------------------------------------------------------------------------------------|------------|------------------|------------------|---------------------|
| Audit_ID                                                                                     | Patient_ID | Old_Patient_Name | New_Patient_Name | Updated_At          |
| 1                                                                                            | P001       | Arun Kumar       | Arun Kumar       | 2026-08-19 11:37:49 |
| NULL                                                                                         | NULL       | NULL             | NULL             | NULL                |

## RESULT:

Thus the trigger is executed successfully.